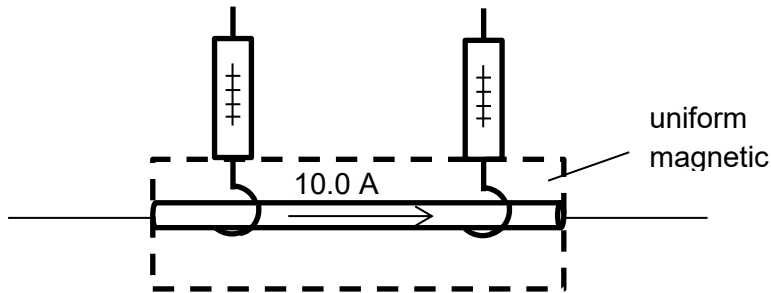


- 24** A 10.0 cm long wire is suspended from two spring balances as shown below.



The wire is placed in a region of uniform magnetic field. The direction of the magnetic field is pointing perpendicularly into the paper and the magnitude of the flux density is $1.00 \times 10^{-3} \text{ T}$. When there is no current in the wire, the reading on each spring balance is 10.00 g.

What is the reading on each of the spring balances when a current of 10.0 A flows through the wire?

- A** 9.90 g **B** 9.95 g **C** 10.05 g **D** 10.10 g